

Dietary Acrylamide Intake and the Risk of Lymphatic Malignancies: The Netherlands Cohort Study on Diet and Cancer



Background Acrylamide, a probable human carcinogen, is present in many everyday foods. Since the finding of its presence in foods in 2002, epidemiological studies have found some suggestive associations between dietary acrylamide exposure and the risk of various cancers. The aim of this prospective study is to investigate for the first time the association between dietary acrylamide intake and the risk of several histological subtypes of lymphatic malignancies. Methods The Netherlands Cohort Study on diet and cancer includes 120,852 men and women followed-up since September 1986. The number of person years at risk was estimated by using a random sample of participants from the total cohort that was chosen at baseline (n = 5,000). Acrylamide intake was estimated from a food frequency questionnaire combined with acrylamide data for Dutch foods. Hazard ratios (HRs) were calculated for acrylamide intake as a continuous variable as well as in categories (quintiles and tertiles), for men and women separately and for never-smokers, using multivariable-adjusted Cox proportional hazards models. Results After 16.3 years of follow-up, 1,233 microscopically confirmed cases of lymphatic malignancies were available for multivariable-adjusted analysis. For multiple myeloma and follicular lymphoma, HRs for men were 1.14 (95% CI: 1.01, 1.27) and 1.28 (95% CI: 1.03, 1.61) per 10 µg acrylamide/day increment, respectively. For never-smoking men, the HR for multiple myeloma was 1.98 (95% CI: 1.38, 2.85). No associations were observed for women. Conclusion We found indications that acrylamide may increase the risk of multiple myeloma and follicular lymphoma in men. This is the first epidemiological study to investigate the association between dietary acrylamide intake and the risk of lymphatic malignancies, and more research into these observed associations is warranted.